$$C_{qd}^{(8)}\text{i1\_i2\_i3\_i4\_} \to$$

$$-\frac{1}{m_\Phi^2} s_\gamma^2 \overline{y_d}^{i2i3} y_d^{i1i4} + \frac{1}{16\pi^2}\Bigg(\frac{1}{72}\frac{1}{m_\Phi^2}\Big(s_\gamma^2\Big(-27\overline{y_d}^{pr}\overline{y_d}^{i2i3}\Big(y_d^{pi4}y_d^{i1r}\big(1+s_\gamma^2\big)-8c_\gamma^2 y_d^{pr}y_d^{i1i4}\Big)+$$
$$9\overline{y_d}^{i2i3}\Big(2y_d^{i1i4}\big(g_1^2-3\big(g_2^2+4g_3^2\big)\big)+y_d^{pi4}\overline{y_u}^{pr}y_u^{i1r}\big(-1+c_\gamma^2\big)\Big)+$$
$$\overline{y_d}^{pi3}\Big(9y_d^{i1i4}\Big(-3\overline{y_d}^{i2r}y_d^{pr}\big(1+s_\gamma^2\big)+\overline{y_u}^{i2r}y_u^{pr}\big(-1+c_\gamma^2\big)\Big)-16g_3^2y_d^{pi4}\delta_{i1i2}\Big)\Big)-$$
$$8g_3^2\Big(s_\gamma^2\overline{y_d}^{i2p}y_d^{i1p}+c_\gamma^2\overline{y_u}^{i2p}y_u^{i1p}\Big)\delta_{i3i4}\Big)+\frac{2}{3}g_3^4\,\mathrm{LF}_{3,0}[m_3]\,\delta_{i1i2}\delta_{i3i4}+$$
$$g_3^4\,\mathrm{LF}_{4,-1}[m_3]\,\delta_{i1i2}\delta_{i3i4}-\frac{16}{15}g_3^4\,\mathrm{LF}_{5,-2}[m_3]\,\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{2}\sum_p s_\gamma g_1^2\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}\left(2s_{2\gamma}c_\gamma+s_\gamma c_{2\gamma}\right)\mathrm{LF}_{1,0}\left[m_{\tilde d}^{p}\right]+$$
$$\frac{2}{9}\sum_p g_3^4\,\mathrm{LF}_{3,0}\left[m_{\tilde d}^{p}\right]\delta_{i1i2}\delta_{i3i4}-\frac{5}{12}\sum_p g_3^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde d}^{p}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{8}{45}\sum_p g_3^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde d}^{p}\right]\delta_{i1i2}\delta_{i3i4}-3s_\gamma\frac{1}{m_\Phi^4}\overline{y_d}^{pr}\overline{y_d}^{i2i3}y_d^{pr}y_d^{i1i4}\left(-s_{2\gamma}c_\gamma+s_\gamma^3\right)\mathrm{LF}_{1,0}\left[m_{\tilde d}^{r}\right]-$$
$$\frac{1}{2}\sum_p s_\gamma g_1^2\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}\left(2s_{2\gamma}c_\gamma+s_\gamma c_{2\gamma}\right)\mathrm{LF}_{1,0}\left[m_{\tilde e}^{p}\right]+$$
$$s_\gamma\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}\overline{y_e}^{pr}y_e^{pr}\left(s_{2\gamma}c_\gamma-s_\gamma^3\right)\mathrm{LF}_{1,0}\left[m_{\tilde e}^{r}\right]+$$
$$\frac{1}{2}s_\gamma\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}\Big(2\overline{y_e}^{pr}y_e^{pr}\left(s_{2\gamma}c_\gamma-s_\gamma^3\right)+\sum_p g_1^2\left(2s_{2\gamma}c_\gamma+s_\gamma c_{2\gamma}\right)\Big)\mathrm{LF}_{1,0}\left[m_{\tilde l}^{p}\right]-$$
$$\frac{1}{2}s_\gamma\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}\Big(6\overline{y_d}^{pr}y_d^{pr}\left(-s_{2\gamma}c_\gamma+s_\gamma^3\right)+6c_\gamma\overline{y_u}^{pr}y_u^{pr}\left(s_{2\gamma}+s_\gamma c_\gamma\right)+$$
$$\sum_p g_1^2\left(2s_{2\gamma}c_\gamma+s_\gamma c_{2\gamma}\right)\Big)\mathrm{LF}_{1,0}\left[m_{\tilde q}^{p}\right]+\frac{4}{9}\sum_p g_3^4\,\mathrm{LF}_{3,0}\left[m_{\tilde q}^{p}\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{5}{6}\sum_p g_3^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde q}^{p}\right]\delta_{i1i2}\delta_{i3i4}+\frac{16}{45}\sum_p g_3^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde q}^{p}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\sum_p s_\gamma g_1^2\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}\left(2s_{2\gamma}c_\gamma+s_\gamma c_{2\gamma}\right)\mathrm{LF}_{1,0}\left[m_{\tilde u}^{p}\right]+\frac{2}{9}\sum_p g_3^4\,\mathrm{LF}_{3,0}\left[m_{\tilde u}^{p}\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{5}{12}\sum_p g_3^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde u}^{p}\right]\delta_{i1i2}\delta_{i3i4}+\frac{8}{45}\sum_p g_3^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde u}^{p}\right]\delta_{i1i2}\delta_{i3i4}-$$
$$3s_\gamma c_\gamma\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}\overline{y_u}^{pr}y_u^{pr}\left(s_{2\gamma}+s_\gamma c_\gamma\right)\mathrm{LF}_{1,0}\left[m_{\tilde u}^{r}\right]-\frac{1}{4}s_\gamma\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}$$
$$\Big(3s_{4\gamma}c_\gamma\left(g_1^2+g_2^2\right)+s_\gamma\Big(g_1^2\left(-1+3c_{2\gamma}^2\right)+3g_2^2\left(-1+c_{2\gamma}^2\right)\Big)\Big)\mathrm{LF}_{1,0}[m_\Phi]+$$
$$\frac{1}{4}\frac{1}{m_\Phi^2}s_\gamma^2\Big(3s_\gamma^2\overline{y_d}^{pr}\overline{y_d}^{i2i3}y_d^{pi4}y_d^{i1r}+3\overline{y_d}^{pi3}y_d^{i1i4}\left(s_\gamma^2\overline{y_d}^{i2r}y_d^{pr}-c_\gamma^2\overline{y_u}^{i2r}y_u^{pr}\right)-$$
$$\overline{y_d}^{i2i3}\Big(2y_d^{i1i4}\left(g_1^2+3g_2^2\right)+3c_\gamma^2y_d^{pi4}\overline{y_u}^{pr}y_u^{i1r}\Big)\Big)\mathrm{LF}_{1,1}[m_\Phi]+$$
$$\frac{1}{12}\Big(s_\gamma^2\Big(3\overline{y_d}^{i2i3}\Big(y_d^{i1i4}\left(g_1^2+3g_2^2\right)+2c_\gamma^2y_d^{pi4}\overline{y_u}^{pr}y_u^{i1r}\Big)+2\overline{y_d}^{pi3}\Big(3c_\gamma^2y_d^{i1i4}\overline{y_u}^{i2r}y_u^{pr}-$$
$$4g_3^2y_d^{pi4}\delta_{i1i2}\Big)\Big)-4g_3^2\left(s_\gamma^2\overline{y_d}^{i2p}y_d^{i1p}+c_\gamma^2\overline{y_u}^{i2p}y_u^{i1p}\right)\delta_{i3i4}\Big)\mathrm{LF}_{1,2}[m_\Phi]+$$
$$\frac{1}{9}g_1^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,0}\left[m_1,m_{\tilde d}^{i3}\right]-\frac{1}{18}g_1^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,1,-1}\left[m_1,m_{\tilde d}^{i3}\right]+$$
$$\frac{1}{54}g_1^2g_3^2\mathrm{LF}_{2,1,0}\left[m_1,m_{\tilde d}^{i3}\right]\delta_{i1i2}\delta_{i3i4}+\frac{1}{54}g_1^2g_3^2\mathrm{LF}_{2,2,-1}\left[m_1,m_{\tilde d}^{i3}\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{27}g_1^2g_3^2\mathrm{LF}_{3,1,-1}\left[m_1,m_{\tilde d}^{i3}\right]\delta_{i1i2}\delta_{i3i4}+\frac{1}{54}g_1^2g_3^2\mathrm{LF}_{4,1,-2}\left[m_1,m_{\tilde d}^{i3}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{9}g_1^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,0}\left[m_1,m_{\tilde d}^{i4}\right]-\frac{1}{18}g_1^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,1,-1}\left[m_1,m_{\tilde d}^{i4}\right]+$$
$$\frac{1}{54}g_1^2g_3^2\mathrm{LF}_{2,1,0}\left[m_1,m_{\tilde d}^{i4}\right]\delta_{i1i2}\delta_{i3i4}+\frac{1}{54}g_1^2g_3^2\mathrm{LF}_{2,2,-1}\left[m_1,m_{\tilde d}^{i4}\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{27}g_1^2g_3^2\mathrm{LF}_{3,1,-1}\left[m_1,m_{\tilde d}^{i4}\right]\delta_{i1i2}\delta_{i3i4}+\frac{1}{54}g_1^2g_3^2\mathrm{LF}_{4,1,-2}\left[m_1,m_{\tilde d}^{i4}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{36}g_1^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,0}\left[m_1,m_{\tilde q}^{i1}\right]-$$
$$\frac{1}{72}g_1^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,1,-1}\left[m_1,m_{\tilde q}^{i1}\right]+\frac{1}{216}g_1^2g_3^2\mathrm{LF}_{2,1,0}\left[m_1,m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{216}g_1^2g_3^2\mathrm{LF}_{2,2,-1}\left[m_1,m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}-\frac{1}{108}g_1^2g_3^2\mathrm{LF}_{3,1,-1}\left[m_1,m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{216}g_1^2g_3^2\mathrm{LF}_{4,1,-2}\left[m_1,m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}+\frac{1}{36}g_1^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,0}\left[m_1,m_{\tilde q}^{i2}\right]-$$
$$\frac{1}{72}g_1^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,1,-1}\left[m_1,m_{\tilde q}^{i2}\right]+\frac{1}{216}g_1^2g_3^2\mathrm{LF}_{2,1,0}\left[m_1,m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{216}g_1^2g_3^2\mathrm{LF}_{2,2,-1}\left[m_1,m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}-\frac{1}{108}g_1^2g_3^2\mathrm{LF}_{3,1,-1}\left[m_1,m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{216}g_1^2g_3^2\mathrm{LF}_{4,1,-2}\left[m_1,m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}+g_1^2\frac{1}{m_\Phi^4}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,-1}[m_1,\tilde\mu]+$$
$$2m_1s_\gamma\tilde\mu c_\gamma g_1^2\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}\left(c_\gamma^2-2s_\gamma^2\right)\mathrm{LF}_{1,1,0}[m_1,\tilde\mu]+$$
$$\frac{3}{4}g_2^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,0}\left[m_2,m_{\tilde q}^{i1}\right]-$$
$$\frac{3}{8}g_2^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,1,-1}\left[m_2,m_{\tilde q}^{i1}\right]+\frac{1}{8}g_2^2g_3^2\mathrm{LF}_{2,1,0}\left[m_2,m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{8}g_2^2g_3^2\mathrm{LF}_{2,2,-1}\left[m_2,m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}-\frac{1}{4}g_2^2g_3^2\mathrm{LF}_{3,1,-1}\left[m_2,m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{8}g_2^2g_3^2\mathrm{LF}_{4,1,-2}\left[m_2,m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}+\frac{3}{4}g_2^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,0}\left[m_2,m_{\tilde q}^{i2}\right]-$$
$$\frac{3}{8}g_2^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,1,-1}\left[m_2,m_{\tilde q}^{i2}\right]+\frac{1}{8}g_2^2g_3^2\mathrm{LF}_{2,1,0}\left[m_2,m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{8}g_2^2g_3^2\mathrm{LF}_{2,2,-1}\left[m_2,m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}-\frac{1}{4}g_2^2g_3^2\mathrm{LF}_{3,1,-1}\left[m_2,m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{8}g_2^2g_3^2\mathrm{LF}_{4,1,-2}\left[m_2,m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}+3g_2^2\frac{1}{m_\Phi^4}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,-1}[m_2,\tilde\mu]+$$
$$6m_2s_\gamma\tilde\mu c_\gamma g_2^2\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}\left(c_\gamma^2-2s_\gamma^2\right)\mathrm{LF}_{1,1,0}[m_2,\tilde\mu]+$$
$$\frac{4}{3}g_3^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,0}\left[m_3,m_{\tilde d}^{i3}\right]-$$
$$\frac{2}{3}g_3^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,1,-1}\left[m_3,m_{\tilde d}^{i3}\right]-\frac{1}{36}g_3^4\mathrm{LF}_{2,1,0}\left[m_3,m_{\tilde d}^{i3}\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{36}g_3^4\mathrm{LF}_{2,2,-1}\left[m_3,m_{\tilde d}^{i3}\right]\delta_{i1i2}\delta_{i3i4}-\frac{25}{36}g_3^4\mathrm{LF}_{3,1,-1}\left[m_3,m_{\tilde d}^{i3}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{2}{9}g_3^4\mathrm{LF}_{4,1,-2}\left[m_3,m_{\tilde d}^{i3}\right]\delta_{i1i2}\delta_{i3i4}+\frac{4}{3}g_3^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,0}\left[m_3,m_{\tilde d}^{i4}\right]-$$
$$\frac{2}{3}g_3^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,1,-1}\left[m_3,m_{\tilde d}^{i4}\right]-\frac{1}{36}g_3^4\mathrm{LF}_{2,1,0}\left[m_3,m_{\tilde d}^{i4}\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{36}g_3^4\mathrm{LF}_{2,2,-1}\left[m_3,m_{\tilde d}^{i4}\right]\delta_{i1i2}\delta_{i3i4}-\frac{25}{36}g_3^4\mathrm{LF}_{3,1,-1}\left[m_3,m_{\tilde d}^{i4}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{2}{9}g_3^4\mathrm{LF}_{4,1,-2}\left[m_3,m_{\tilde d}^{i4}\right]\delta_{i1i2}\delta_{i3i4}+\frac{4}{3}g_3^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,0}\left[m_3,m_{\tilde q}^{i1}\right]-$$
$$\frac{2}{3}g_3^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,1,-1}\left[m_3,m_{\tilde q}^{i1}\right]-\frac{1}{36}g_3^4\mathrm{LF}_{2,1,0}\left[m_3,m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{36}g_3^4\mathrm{LF}_{2,2,-1}\left[m_3,m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}-\frac{25}{36}g_3^4\mathrm{LF}_{3,1,-1}\left[m_3,m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{2}{9}g_3^4\mathrm{LF}_{4,1,-2}\left[m_3,m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}+\frac{4}{3}g_3^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,0}\left[m_3,m_{\tilde q}^{i2}\right]-$$
$$\frac{2}{3}g_3^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,1,-1}\left[m_3,m_{\tilde q}^{i2}\right]-\frac{1}{36}g_3^4\mathrm{LF}_{2,1,0}\left[m_3,m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{36}g_3^4\mathrm{LF}_{2,2,-1}\left[m_3,m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}-\frac{25}{36}g_3^4\mathrm{LF}_{3,1,-1}\left[m_3,m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{2}{9}g_3^4\mathrm{LF}_{4,1,-2}\left[m_3,m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}-\frac{1}{3}g_3^2\overline{y_d}^{i2p}y_d^{i1p}\mathrm{LF}_{2,1,0}\left[m_{\tilde d}^{p},\tilde\mu\right]\delta_{i3i4}+$$
$$\frac{1}{6}g_3^2\overline{y_d}^{i2p}y_d^{i1p}\mathrm{LF}_{2,2,-1}\left[m_{\tilde d}^{p},\tilde\mu\right]\delta_{i3i4}+\frac{1}{6}g_3^2\overline{y_d}^{i2p}y_d^{i1p}\mathrm{LF}_{3,1,-1}\left[m_{\tilde d}^{p},\tilde\mu\right]\delta_{i3i4}+$$
$$3s_\gamma\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}\Big(\tilde\mu c_\gamma\overline{y_d}^{pr}\Big(a_d^{pr}\left(c_\gamma^2-2s_\gamma^2\right)-3s_\gamma\tilde\mu c_\gamma y_d^{pr}\Big)+$$
$$\overline{a_d}^{pr}\Big(a_d^{pr}\left(2s_\gamma c_\gamma^2-s_\gamma^3\right)+\tilde\mu c_\gamma y_d^{pr}\left(c_\gamma^2-2s_\gamma^2\right)\Big)\Big)\mathrm{LF}_{1,1,0}\left[m_{\tilde d}^{r},m_{\tilde q}^{p}\right]+$$
$$\frac{1}{2}\frac{1}{m_\Phi^2}s_\gamma^2\Big(\overline{y_d}^{pr}\overline{y_d}^{i2i3}y_d^{pi4}y_d^{i1r}+\overline{y_d}^{pi3}\overline{y_d}^{i2r}y_d^{pr}y_d^{i1i4}\Big)\mathrm{LF}_{1,1,0}\left[m_{\tilde d}^{r},\tilde\mu\right]-$$
$$\frac{1}{27}g_1^2g_3^2\mathrm{LF}_{2,1,0}\left[m_{\tilde d}^{i3},m_1\right]\delta_{i1i2}\delta_{i3i4}+\frac{1}{54}g_1^2g_3^2\mathrm{LF}_{3,1,-1}\left[m_{\tilde d}^{i3},m_1\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{18}g_3^4\mathrm{LF}_{2,1,0}\left[m_{\tilde d}^{i3},m_3\right]\delta_{i1i2}\delta_{i3i4}-\frac{1}{36}g_3^4\mathrm{LF}_{3,1,-1}\left[m_{\tilde d}^{i3},m_3\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{27}g_1^2g_3^2\mathrm{LF}_{2,1,0}\left[m_{\tilde d}^{i4},m_1\right]\delta_{i1i2}\delta_{i3i4}+\frac{1}{54}g_1^2g_3^2\mathrm{LF}_{3,1,-1}\left[m_{\tilde d}^{i4},m_1\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{18}g_3^4\mathrm{LF}_{2,1,0}\left[m_{\tilde d}^{i4},m_3\right]\delta_{i1i2}\delta_{i3i4}-\frac{1}{36}g_3^4\mathrm{LF}_{3,1,-1}\left[m_{\tilde d}^{i4},m_3\right]\delta_{i1i2}\delta_{i3i4}+$$
$$s_\gamma\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}\Big(\tilde\mu c_\gamma\overline{y_e}^{pr}\Big(a_e^{pr}\left(c_\gamma^2-2s_\gamma^2\right)-3s_\gamma\tilde\mu c_\gamma y_e^{pr}\Big)+$$
$$\overline{a_e}^{pr}\Big(a_e^{pr}\left(2s_\gamma c_\gamma^2-s_\gamma^3\right)+\tilde\mu c_\gamma y_e^{pr}\left(c_\gamma^2-2s_\gamma^2\right)\Big)\Big)\mathrm{LF}_{1,1,0}\left[m_{\tilde e}^{r},m_{\tilde l}^{p}\right]+s_\gamma c_\gamma\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}$$
$$y_d^{i1i4}\Big(\tilde\mu\overline{y_e}^{pr}\Big(a_e^{pr}\left(-c_\gamma^2+s_\gamma^2\right)+2s_\gamma\tilde\mu c_\gamma y_e^{pr}\Big)+\overline{a_e}^{pr}\Big(-2s_\gamma c_\gamma a_e^{pr}+\tilde\mu y_e^{pr}\left(-c_\gamma^2+s_\gamma^2\right)\Big)\Big)$$
$$\mathrm{LF}_{2,1,0}\left[m_{\tilde l}^{p},m_{\tilde e}^{r}\right]+s_\gamma c_\gamma\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\Big(\tilde\mu\overline{y_e}^{pr}\Big(a_e^{pr}\left(c_\gamma^2-s_\gamma^2\right)-2s_\gamma\tilde\mu c_\gamma y_e^{pr}\Big)+$$
$$\overline{a_e}^{pr}\Big(2s_\gamma c_\gamma a_e^{pr}+\tilde\mu y_e^{pr}\left(c_\gamma^2-s_\gamma^2\right)\Big)\Big)\mathrm{LF}_{3,1,-1}\left[m_{\tilde l}^{p},m_{\tilde e}^{r}\right]-$$
$$c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\left(c_\gamma\overline{a_e}^{pr}-s_\gamma\tilde\mu\overline{y_e}^{pr}\right)\left(c_\gamma a_e^{pr}-s_\gamma\tilde\mu y_e^{pr}\right)\mathrm{LF}_{3,1,0}\left[m_{\tilde l}^{p},m_{\tilde e}^{r}\right]+$$
$$3c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\left(c_\gamma\overline{a_e}^{pr}-s_\gamma\tilde\mu\overline{y_e}^{pr}\right)\left(c_\gamma a_e^{pr}-s_\gamma\tilde\mu y_e^{pr}\right)\mathrm{LF}_{4,1,-1}\left[m_{\tilde l}^{p},m_{\tilde e}^{r}\right]-2c_\gamma^2\overline{y_d}^{i2i3}$$
$$y_d^{i1i4}\left(c_\gamma\overline{a_e}^{pr}-s_\gamma\tilde\mu\overline{y_e}^{pr}\right)\left(c_\gamma a_e^{pr}-s_\gamma\tilde\mu y_e^{pr}\right)\mathrm{LF}_{5,1,-2}\left[m_{\tilde l}^{p},m_{\tilde e}^{r}\right]+3s_\gamma c_\gamma\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}$$
$$\Big(\tilde\mu\overline{y_d}^{pr}\Big(a_d^{pr}\left(-c_\gamma^2+s_\gamma^2\right)+2s_\gamma\tilde\mu c_\gamma y_d^{pr}\Big)+\overline{a_d}^{pr}\Big(-2s_\gamma c_\gamma a_d^{pr}+\tilde\mu y_d^{pr}\left(-c_\gamma^2+s_\gamma^2\right)\Big)\Big)$$
$$\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{p},m_{\tilde d}^{r}\right]+3s_\gamma c_\gamma\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\Big(\tilde\mu\overline{y_d}^{pr}\Big(a_d^{pr}\left(c_\gamma^2-s_\gamma^2\right)-2s_\gamma\tilde\mu c_\gamma y_d^{pr}\Big)+$$
$$\overline{a_d}^{pr}\Big(2s_\gamma c_\gamma a_d^{pr}+\tilde\mu y_d^{pr}\left(c_\gamma^2-s_\gamma^2\right)\Big)\Big)\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{p},m_{\tilde d}^{r}\right]-$$
$$3c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\left(c_\gamma\overline{a_d}^{pr}-s_\gamma\tilde\mu\overline{y_d}^{pr}\right)\left(c_\gamma a_d^{pr}-s_\gamma\tilde\mu y_d^{pr}\right)\mathrm{LF}_{3,1,0}\left[m_{\tilde q}^{p},m_{\tilde d}^{r}\right]+$$
$$9c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\left(c_\gamma\overline{a_d}^{pr}-s_\gamma\tilde\mu\overline{y_d}^{pr}\right)\left(c_\gamma a_d^{pr}-s_\gamma\tilde\mu y_d^{pr}\right)\mathrm{LF}_{4,1,-1}\left[m_{\tilde q}^{p},m_{\tilde d}^{r}\right]-$$
$$6c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\left(c_\gamma\overline{a_d}^{pr}-s_\gamma\tilde\mu\overline{y_d}^{pr}\right)\left(c_\gamma a_d^{pr}-s_\gamma\tilde\mu y_d^{pr}\right)\mathrm{LF}_{5,1,-2}\left[m_{\tilde q}^{p},m_{\tilde d}^{r}\right]+$$
$$3s_\gamma\frac{1}{m_\Phi^4}\overline{y_d}^{i2i3}y_d^{i1i4}\Big(c_\gamma\overline{a_u}^{pr}\Big(-3s_\gamma c_\gamma a_u^{pr}+\tilde\mu y_u^{pr}\left(c_\gamma^2-2s_\gamma^2\right)\Big)+$$
$$\tilde\mu\overline{y_u}^{pr}\Big(a_u^{pr}\left(c_\gamma^3-2c_\gamma s_\gamma^2\right)-s_\gamma\tilde\mu y_u^{pr}\left(-2c_\gamma^2+s_\gamma^2\right)\Big)\Big)\mathrm{LF}_{1,1,0}\left[m_{\tilde q}^{p},m_{\tilde u}^{r}\right]+$$
$$\frac{1}{m_\Phi^2}s_\gamma^2\Big(\overline{y_d}^{pr}\overline{y_d}^{i2i3}y_d^{pi4}y_d^{i1r}+\overline{y_d}^{pi3}\overline{y_d}^{i2r}y_d^{pr}y_d^{i1i4}\Big)\mathrm{LF}_{1,1,0}\left[m_{\tilde q}^{p},\tilde\mu\right]-$$
$$\frac{2}{3}g_3^2\overline{y_d}^{pi3}y_d^{pi4}\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{p},\tilde\mu\right]\delta_{i1i2}+\frac{1}{3}g_3^2\overline{y_d}^{pi3}y_d^{pi4}\mathrm{LF}_{2,2,-1}\left[m_{\tilde q}^{p},\tilde\mu\right]\delta_{i1i2}+$$
$$\frac{1}{3}g_3^2\overline{y_d}^{pi3}y_d^{pi4}\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{p},\tilde\mu\right]\delta_{i1i2}-\frac{1}{108}g_1^2g_3^2\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i1},m_1\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{216}g_1^2g_3^2\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i1},m_1\right]\delta_{i1i2}\delta_{i3i4}-\frac{1}{4}g_2^2g_3^2\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i1},m_2\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{8}g_2^2g_3^2\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i1},m_2\right]\delta_{i1i2}\delta_{i3i4}+\frac{1}{18}g_3^4\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i1},m_3\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{36}g_3^4\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i1},m_3\right]\delta_{i1i2}\delta_{i3i4}-\frac{1}{108}g_1^2g_3^2\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i2},m_1\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{216}g_1^2g_3^2\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i2},m_1\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{4}g_2^2g_3^2\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i2},m_2\right]\delta_{i1i2}\delta_{i3i4}+\frac{1}{8}g_2^2g_3^2\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i2},m_2\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{18}g_3^4\mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i2},m_3\right]\delta_{i1i2}\delta_{i3i4}-\frac{1}{36}g_3^4\mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i2},m_3\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{1}{3}g_3^2\overline{y_u}^{i2p}y_u^{i1p}\mathrm{LF}_{2,1,0}\left[m_{\tilde u}^{p},\tilde\mu\right]\delta_{i3i4}+\frac{1}{6}g_3^2\overline{y_u}^{i2p}y_u^{i1p}\mathrm{LF}_{2,2,-1}\left[m_{\tilde u}^{p},\tilde\mu\right]\delta_{i3i4}+$$
$$\frac{1}{6}g_3^2\overline{y_u}^{i2p}y_u^{i1p}\mathrm{LF}_{3,1,-1}\left[m_{\tilde u}^{p},\tilde\mu\right]\delta_{i3i4}+3s_\gamma c_\gamma\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}$$
$$\Big(\tilde\mu\overline{y_u}^{pr}\Big(a_u^{pr}\left(-c_\gamma^2+s_\gamma^2\right)-2s_\gamma\tilde\mu c_\gamma y_u^{pr}\Big)+\overline{a_u}^{pr}\Big(2s_\gamma c_\gamma a_u^{pr}+\tilde\mu y_u^{pr}\left(-c_\gamma^2+s_\gamma^2\right)\Big)\Big)$$
$$\mathrm{LF}_{2,1,0}\left[m_{\tilde u}^{r},m_{\tilde q}^{p}\right]+3s_\gamma c_\gamma\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\Big(\tilde\mu\overline{y_u}^{pr}\Big(a_u^{pr}\left(c_\gamma^2-s_\gamma^2\right)+2s_\gamma\tilde\mu c_\gamma y_u^{pr}\Big)+$$
$$\overline{a_u}^{pr}\Big(-2s_\gamma c_\gamma a_u^{pr}+\tilde\mu y_u^{pr}\left(c_\gamma^2-s_\gamma^2\right)\Big)\Big)\mathrm{LF}_{3,1,-1}\left[m_{\tilde u}^{r},m_{\tilde q}^{p}\right]-$$
$$3c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\left(-s_\gamma\overline{a_u}^{pr}+\tilde\mu c_\gamma\overline{y_u}^{pr}\right)\left(-s_\gamma a_u^{pr}+\tilde\mu c_\gamma y_u^{pr}\right)\mathrm{LF}_{3,1,0}\left[m_{\tilde u}^{r},m_{\tilde q}^{p}\right]+$$
$$9c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\left(-s_\gamma\overline{a_u}^{pr}+\tilde\mu c_\gamma\overline{y_u}^{pr}\right)\left(-s_\gamma a_u^{pr}+\tilde\mu c_\gamma y_u^{pr}\right)\mathrm{LF}_{4,1,-1}\left[m_{\tilde u}^{r},m_{\tilde q}^{p}\right]-$$
$$6c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\left(-s_\gamma\overline{a_u}^{pr}+\tilde\mu c_\gamma\overline{y_u}^{pr}\right)\left(-s_\gamma a_u^{pr}+\tilde\mu c_\gamma y_u^{pr}\right)\mathrm{LF}_{5,1,-2}\left[m_{\tilde u}^{r},m_{\tilde q}^{p}\right]+$$
$$\frac{1}{2}\frac{1}{m_\Phi^2}s_\gamma^2\Big(\overline{y_d}^{pi3}y_d^{i1i4}\overline{y_u}^{i2r}y_u^{pr}+\overline{y_d}^{i2i3}y_d^{pi4}\overline{y_u}^{pr}y_u^{i1r}\Big)\mathrm{LF}_{1,1,0}\left[m_{\tilde u}^{r},\tilde\mu\right]+$$
$$2m_1s_\gamma\tilde\mu c_\gamma g_1^2\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\left(-c_\gamma^2+s_\gamma^2\right)\mathrm{LF}_{2,1,0}[\tilde\mu,m_1]+$$
$$2c_\gamma g_1^2\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\Big(c_\gamma m_\Phi^2+m_1s_\gamma\tilde\mu\left(c_\gamma^2-s_\gamma^2\right)\Big)\mathrm{LF}_{3,1,-1}[\tilde\mu,m_1]+$$
$$2m_1s_\gamma\tilde\mu g_1^2c_\gamma^3\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{3,1,0}[\tilde\mu,m_1]-4g_1^2c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{4,1,-2}[\tilde\mu,m_1]-$$
$$6m_1s_\gamma\tilde\mu g_1^2c_\gamma^3\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{4,1,-1}[\tilde\mu,m_1]+$$
$$2g_1^2c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{5,1,-3}[\tilde\mu,m_1]+4m_1s_\gamma\tilde\mu g_1^2c_\gamma^3\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{5,1,-2}[\tilde\mu,m_1]+$$
$$6m_2s_\gamma\tilde\mu c_\gamma g_2^2\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\left(-c_\gamma^2+s_\gamma^2\right)\mathrm{LF}_{2,1,0}[\tilde\mu,m_2]+$$
$$6c_\gamma g_2^2\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\Big(c_\gamma m_\Phi^2+m_2s_\gamma\tilde\mu\left(c_\gamma^2-s_\gamma^2\right)\Big)\mathrm{LF}_{3,1,-1}[\tilde\mu,m_2]+$$
$$6m_2s_\gamma\tilde\mu g_2^2c_\gamma^3\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{3,1,0}[\tilde\mu,m_2]-12g_2^2c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{4,1,-2}[\tilde\mu,m_2]-$$
$$18m_2s_\gamma\tilde\mu g_2^2c_\gamma^3\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{4,1,-1}[\tilde\mu,m_2]+6g_2^2c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{5,1,-3}[\tilde\mu,m_2]+$$
$$12m_2s_\gamma\tilde\mu g_2^2c_\gamma^3\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{5,1,-2}[\tilde\mu,m_2]+\frac{1}{6}g_3^2\overline{y_d}^{i2p}y_d^{i1p}\mathrm{LF}_{2,1,0}\left[\tilde\mu,m_{\tilde d}^{p}\right]\delta_{i3i4}-$$
$$\frac{1}{3}g_3^2\overline{y_d}^{i2p}y_d^{i1p}\mathrm{LF}_{3,1,-1}\left[\tilde\mu,m_{\tilde d}^{p}\right]\delta_{i3i4}+\frac{1}{6}g_3^2\overline{y_d}^{i2p}y_d^{i1p}\mathrm{LF}_{4,1,-2}\left[\tilde\mu,m_{\tilde d}^{p}\right]\delta_{i3i4}-$$
$$\frac{1}{4}\frac{1}{m_\Phi^2}s_\gamma^2\Big(\overline{y_d}^{pr}\overline{y_d}^{i2i3}y_d^{pi4}y_d^{i1r}+\overline{y_d}^{pi3}\overline{y_d}^{i2r}y_d^{pr}y_d^{i1i4}\Big)\mathrm{LF}_{2,1,-1}\left[\tilde\mu,m_{\tilde d}^{r}\right]-$$
$$\frac{1}{2}\frac{1}{m_\Phi^2}s_\gamma^2\Big(\overline{y_d}^{pr}\overline{y_d}^{i2i3}y_d^{pi4}y_d^{i1r}+\overline{y_d}^{pi3}\overline{y_d}^{i2r}y_d^{pr}y_d^{i1i4}\Big)\mathrm{LF}_{2,1,-1}\left[\tilde\mu,m_{\tilde q}^{p}\right]+$$
$$\frac{1}{3}g_3^2\overline{y_d}^{pi3}y_d^{pi4}\mathrm{LF}_{2,1,0}\left[\tilde\mu,m_{\tilde q}^{p}\right]\delta_{i1i2}-\frac{2}{3}g_3^2\overline{y_d}^{pi3}y_d^{pi4}\mathrm{LF}_{3,1,-1}\left[\tilde\mu,m_{\tilde q}^{p}\right]\delta_{i1i2}+$$
$$\frac{1}{3}g_3^2\overline{y_d}^{pi3}y_d^{pi4}\mathrm{LF}_{4,1,-2}\left[\tilde\mu,m_{\tilde q}^{p}\right]\delta_{i1i2}+\frac{1}{6}g_3^2\overline{y_u}^{i2p}y_u^{i1p}\mathrm{LF}_{2,1,0}\left[\tilde\mu,m_{\tilde u}^{p}\right]\delta_{i3i4}-$$
$$\frac{1}{3}g_3^2\overline{y_u}^{i2p}y_u^{i1p}\mathrm{LF}_{3,1,-1}\left[\tilde\mu,m_{\tilde u}^{p}\right]\delta_{i3i4}+\frac{1}{6}g_3^2\overline{y_u}^{i2p}y_u^{i1p}\mathrm{LF}_{4,1,-2}\left[\tilde\mu,m_{\tilde u}^{p}\right]\delta_{i3i4}-$$
$$\frac{1}{4}\frac{1}{m_\Phi^2}s_\gamma^2\Big(\overline{y_d}^{pi3}y_d^{i1i4}\overline{y_u}^{i2r}y_u^{pr}+\overline{y_d}^{i2i3}y_d^{pi4}\overline{y_u}^{pr}y_u^{i1r}\Big)\mathrm{LF}_{2,1,-1}\left[\tilde\mu,m_{\tilde u}^{r}\right]+$$
$$\frac{1}{9}m_1s_\gamma g_1^2\frac{1}{m_\Phi^2}y_d^{i1i4}\left(s_\gamma\overline{a_d}^{i2i3}+\tilde\mu c_\gamma\overline{y_d}^{i2i3}\right)\mathrm{LF}_{1,1,1,0}\left[m_1,m_{\tilde d}^{i3},m_{\tilde q}^{i2}\right]+$$
$$\frac{1}{36}m_1c_\gamma g_1^2y_d^{i1i4}\left(-c_\gamma\overline{a_d}^{i2i3}+s_\gamma\tilde\mu\overline{y_d}^{i2i3}\right)\mathrm{LF}_{2,2,1,-1}\left[m_1,m_{\tilde d}^{i3},m_{\tilde q}^{i2}\right]-$$
$$\frac{1}{3}g_1^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,-1}\left[m_1,m_{\tilde d}^{i3},\tilde\mu\right]+$$
$$\frac{1}{3}m_1s_\gamma\tilde\mu c_\gamma g_1^2\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,0}\left[m_1,m_{\tilde d}^{i3},\tilde\mu\right]+$$
$$\frac{1}{9}m_1s_\gamma g_1^2\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}\left(s_\gamma a_d^{i1i4}+\tilde\mu c_\gamma y_d^{i1i4}\right)\mathrm{LF}_{1,1,1,0}\left[m_1,m_{\tilde d}^{i4},m_{\tilde q}^{i1}\right]+$$
$$\frac{1}{36}m_1c_\gamma g_1^2\overline{y_d}^{i2i3}\left(-c_\gamma a_d^{i1i4}+s_\gamma\tilde\mu y_d^{i1i4}\right)\mathrm{LF}_{2,2,1,-1}\left[m_1,m_{\tilde d}^{i4},m_{\tilde q}^{i1}\right]-$$
$$\frac{1}{3}g_1^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,-1}\left[m_1,m_{\tilde d}^{i4},\tilde\mu\right]+$$
$$\frac{1}{3}m_1s_\gamma\tilde\mu c_\gamma g_1^2\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,0}\left[m_1,m_{\tilde d}^{i4},\tilde\mu\right]+$$
$$\frac{1}{36}m_1c_\gamma g_1^2\overline{y_d}^{i2i3}\left(-c_\gamma a_d^{i1i4}+s_\gamma\tilde\mu y_d^{i1i4}\right)\mathrm{LF}_{2,2,1,-1}\left[m_1,m_{\tilde q}^{i1},m_{\tilde d}^{i4}\right]-$$
$$\frac{1}{6}g_1^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,-1}\left[m_1,m_{\tilde q}^{i1},\tilde\mu\right]+$$
$$\frac{1}{6}m_1s_\gamma\tilde\mu c_\gamma g_1^2\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,0}\left[m_1,m_{\tilde q}^{i1},\tilde\mu\right]+$$
$$\frac{1}{36}m_1c_\gamma g_1^2y_d^{i1i4}\left(-c_\gamma\overline{a_d}^{i2i3}+s_\gamma\tilde\mu\overline{y_d}^{i2i3}\right)\mathrm{LF}_{2,2,1,-1}\left[m_1,m_{\tilde q}^{i2},m_{\tilde d}^{i3}\right]-$$
$$\frac{1}{6}g_1^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,-1}\left[m_1,m_{\tilde q}^{i2},\tilde\mu\right]+$$
$$\frac{1}{6}m_1s_\gamma\tilde\mu c_\gamma g_1^2\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,0}\left[m_1,m_{\tilde q}^{i2},\tilde\mu\right]-$$
$$\frac{1}{6}g_1^2c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,2,1,-2}\left[m_1,\tilde\mu,m_{\tilde d}^{i3}\right]-$$
$$\frac{1}{6}m_1s_\gamma\tilde\mu c_\gamma g_1^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,2,1,-1}\left[m_1,\tilde\mu,m_{\tilde d}^{i3}\right]-$$
$$\frac{1}{6}g_1^2c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,2,1,-2}\left[m_1,\tilde\mu,m_{\tilde d}^{i4}\right]-$$
$$\frac{1}{6}m_1s_\gamma\tilde\mu c_\gamma g_1^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,2,1,-1}\left[m_1,\tilde\mu,m_{\tilde d}^{i4}\right]-$$
$$\frac{1}{12}g_1^2c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,2,1,-2}\left[m_1,\tilde\mu,m_{\tilde q}^{i1}\right]-$$
$$\frac{1}{12}m_1s_\gamma\tilde\mu c_\gamma g_1^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,2,1,-1}\left[m_1,\tilde\mu,m_{\tilde q}^{i1}\right]-$$
$$\frac{1}{12}g_1^2c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,2,1,-2}\left[m_1,\tilde\mu,m_{\tilde q}^{i2}\right]-$$
$$\frac{1}{12}m_1s_\gamma\tilde\mu c_\gamma g_1^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,2,1,-1}\left[m_1,\tilde\mu,m_{\tilde q}^{i2}\right]-$$
$$\frac{3}{2}g_2^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,-1}\left[m_2,m_{\tilde q}^{i1},\tilde\mu\right]+$$
$$\frac{3}{2}m_2s_\gamma\tilde\mu c_\gamma g_2^2\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,0}\left[m_2,m_{\tilde q}^{i1},\tilde\mu\right]-$$
$$\frac{3}{2}g_2^2\frac{1}{m_\Phi^2}s_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,-1}\left[m_2,m_{\tilde q}^{i2},\tilde\mu\right]+$$
$$\frac{3}{2}m_2s_\gamma\tilde\mu c_\gamma g_2^2\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,0}\left[m_2,m_{\tilde q}^{i2},\tilde\mu\right]-$$
$$\frac{3}{4}g_2^2c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,2,1,-2}\left[m_2,\tilde\mu,m_{\tilde q}^{i1}\right]-$$
$$\frac{3}{4}m_2s_\gamma\tilde\mu c_\gamma g_2^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,2,1,-1}\left[m_2,\tilde\mu,m_{\tilde q}^{i1}\right]-$$
$$\frac{3}{4}g_2^2c_\gamma^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,2,1,-2}\left[m_2,\tilde\mu,m_{\tilde q}^{i2}\right]-$$
$$\frac{3}{4}m_2s_\gamma\tilde\mu c_\gamma g_2^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{2,2,1,-1}\left[m_2,\tilde\mu,m_{\tilde q}^{i2}\right]+$$
$$\frac{1}{6}g_3^4\mathrm{LF}_{2,1,1,-1}\left[m_3,m_{\tilde d}^{i3},m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}-\frac{7}{6}g_3^4m_3^2\mathrm{LF}_{2,1,1,0}\left[m_3,m_{\tilde d}^{i3},m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}-$$
$$\frac{8}{3}m_3s_\gamma g_3^2\frac{1}{m_\Phi^2}y_d^{i1i4}\left(s_\gamma\overline{a_d}^{i2i3}+\tilde\mu c_\gamma\overline{y_d}^{i2i3}\right)\mathrm{LF}_{1,1,1,0}\left[m_3,m_{\tilde d}^{i3},m_{\tilde q}^{i2}\right]+$$
$$\frac{2}{3}m_3c_\gamma g_3^2y_d^{i1i4}\left(c_\gamma\overline{a_d}^{i2i3}-s_\gamma\tilde\mu\overline{y_d}^{i2i3}\right)\mathrm{LF}_{2,2,1,-1}\left[m_3,m_{\tilde d}^{i3},m_{\tilde q}^{i2}\right]-$$
$$\frac{8}{3}m_3s_\gamma g_3^2\frac{1}{m_\Phi^2}\overline{y_d}^{i2i3}\left(s_\gamma a_d^{i1i4}+\tilde\mu c_\gamma y_d^{i1i4}\right)\mathrm{LF}_{1,1,1,0}\left[m_3,m_{\tilde d}^{i4},m_{\tilde q}^{i1}\right]+$$
$$\frac{2}{3}m_3c_\gamma g_3^2\overline{y_d}^{i2i3}\left(c_\gamma a_d^{i1i4}-s_\gamma\tilde\mu y_d^{i1i4}\right)\mathrm{LF}_{2,2,1,-1}\left[m_3,m_{\tilde d}^{i4},m_{\tilde q}^{i1}\right]+$$
$$\frac{1}{6}g_3^4\mathrm{LF}_{2,1,1,-1}\left[m_3,m_{\tilde d}^{i4},m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}-\frac{7}{6}g_3^4m_3^2\mathrm{LF}_{2,1,1,0}\left[m_3,m_{\tilde d}^{i4},m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{2}{3}m_3c_\gamma g_3^2\overline{y_d}^{i2i3}\left(c_\gamma a_d^{i1i4}-s_\gamma\tilde\mu y_d^{i1i4}\right)\mathrm{LF}_{2,2,1,-1}\left[m_3,m_{\tilde q}^{i1},m_{\tilde d}^{i4}\right]+$$
$$\frac{2}{3}m_3c_\gamma g_3^2y_d^{i1i4}\left(c_\gamma\overline{a_d}^{i2i3}-s_\gamma\tilde\mu\overline{y_d}^{i2i3}\right)\mathrm{LF}_{2,2,1,-1}\left[m_3,m_{\tilde q}^{i2},m_{\tilde d}^{i3}\right]+$$
$$\frac{1}{18}m_1c_\gamma g_1^2y_d^{i1i4}\left(c_\gamma\overline{a_d}^{i2i3}-s_\gamma\tilde\mu\overline{y_d}^{i2i3}\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde d}^{i3},m_1,m_{\tilde q}^{i2}\right]+$$
$$\frac{1}{18}m_1c_\gamma g_1^2y_d^{i1i4}\left(-c_\gamma\overline{a_d}^{i2i3}+s_\gamma\tilde\mu\overline{y_d}^{i2i3}\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde d}^{i3},m_1,m_{\tilde q}^{i2}\right]+$$
$$\frac{4}{3}m_3c_\gamma g_3^2y_d^{i1i4}\left(-c_\gamma\overline{a_d}^{i2i3}+s_\gamma\tilde\mu\overline{y_d}^{i2i3}\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde d}^{i3},m_3,m_{\tilde q}^{i2}\right]+$$
$$\frac{4}{3}m_3c_\gamma g_3^2y_d^{i1i4}\left(c_\gamma\overline{a_d}^{i2i3}-s_\gamma\tilde\mu\overline{y_d}^{i2i3}\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde d}^{i3},m_3,m_{\tilde q}^{i2}\right]+$$
$$\frac{1}{18}m_1c_\gamma g_1^2\overline{y_d}^{i2i3}\left(c_\gamma a_d^{i1i4}-s_\gamma\tilde\mu y_d^{i1i4}\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde d}^{i4},m_1,m_{\tilde q}^{i1}\right]+$$
$$\frac{1}{18}m_1c_\gamma g_1^2\overline{y_d}^{i2i3}\left(-c_\gamma a_d^{i1i4}+s_\gamma\tilde\mu y_d^{i1i4}\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde d}^{i4},m_1,m_{\tilde q}^{i1}\right]+$$
$$\frac{4}{3}m_3c_\gamma g_3^2\overline{y_d}^{i2i3}\left(-c_\gamma a_d^{i1i4}+s_\gamma\tilde\mu y_d^{i1i4}\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde d}^{i4},m_3,m_{\tilde q}^{i1}\right]+$$
$$\frac{4}{3}m_3c_\gamma g_3^2\overline{y_d}^{i2i3}\left(c_\gamma a_d^{i1i4}-s_\gamma\tilde\mu y_d^{i1i4}\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde d}^{i4},m_3,m_{\tilde q}^{i1}\right]-$$
$$s_\gamma\tilde\mu\frac{1}{m_\Phi^2}\Big(\overline{y_d}^{i2i3}y_d^{pi4}y_u^{i1r}\left(c_\gamma\overline{a_u}^{pr}+s_\gamma\tilde\mu\overline{y_u}^{pr}\right)+\overline{y_d}^{pi3}y_d^{i1i4}\overline{y_u}^{i2r}\left(c_\gamma a_u^{pr}+s_\gamma\tilde\mu y_u^{pr}\right)\Big)$$
$$\mathrm{LF}_{1,1,1,0}\left[m_{\tilde q}^{p},m_{\tilde u}^{r},\tilde\mu\right]+\frac{1}{2}\tilde\mu c_\gamma\Big(\overline{y_d}^{i2i3}y_d^{pi4}y_u^{i1r}\left(s_\gamma\overline{a_u}^{pr}-\tilde\mu c_\gamma\overline{y_u}^{pr}\right)+$$
$$\overline{y_d}^{pi3}y_d^{i1i4}\overline{y_u}^{i2r}\left(s_\gamma a_u^{pr}-\tilde\mu c_\gamma y_u^{pr}\right)\Big)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde q}^{p},m_{\tilde u}^{r},\tilde\mu\right]+$$
$$\frac{1}{2}\tilde\mu c_\gamma\Big(\overline{y_d}^{i2i3}y_d^{pi4}y_u^{i1r}\left(-s_\gamma\overline{a_u}^{pr}+\tilde\mu c_\gamma\overline{y_u}^{pr}\right)+\overline{y_d}^{pi3}y_d^{i1i4}\overline{y_u}^{i2r}\left(-s_\gamma a_u^{pr}+\tilde\mu c_\gamma y_u^{pr}\right)\Big)$$
$$\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde q}^{p},m_{\tilde u}^{r},\tilde\mu\right]+\frac{1}{4}\tilde\mu c_\gamma\Big(\overline{y_d}^{i2i3}y_d^{pi4}y_u^{i1r}\left(-s_\gamma\overline{a_u}^{pr}+\tilde\mu c_\gamma\overline{y_u}^{pr}\right)+$$
$$\overline{y_d}^{pi3}y_d^{i1i4}\overline{y_u}^{i2r}\left(-s_\gamma a_u^{pr}+\tilde\mu c_\gamma y_u^{pr}\right)\Big)\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde q}^{p},\tilde\mu,m_{\tilde u}^{r}\right]+$$
$$\frac{1}{18}m_1c_\gamma g_1^2\overline{y_d}^{i2i3}\left(c_\gamma a_d^{i1i4}-s_\gamma\tilde\mu y_d^{i1i4}\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde q}^{i1},m_1,m_{\tilde d}^{i4}\right]+$$
$$\frac{1}{18}m_1c_\gamma g_1^2\overline{y_d}^{i2i3}\left(-c_\gamma a_d^{i1i4}+s_\gamma\tilde\mu y_d^{i1i4}\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde q}^{i1},m_1,m_{\tilde d}^{i4}\right]+$$
$$\frac{4}{3}m_3c_\gamma g_3^2\overline{y_d}^{i2i3}\left(-c_\gamma a_d^{i1i4}+s_\gamma\tilde\mu y_d^{i1i4}\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde q}^{i1},m_3,m_{\tilde d}^{i4}\right]+$$
$$\frac{4}{3}m_3c_\gamma g_3^2\overline{y_d}^{i2i3}\left(c_\gamma a_d^{i1i4}-s_\gamma\tilde\mu y_d^{i1i4}\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde q}^{i1},m_3,m_{\tilde d}^{i4}\right]+$$
$$\frac{1}{18}m_1c_\gamma g_1^2y_d^{i1i4}\left(c_\gamma\overline{a_d}^{i2i3}-s_\gamma\tilde\mu\overline{y_d}^{i2i3}\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde q}^{i2},m_1,m_{\tilde d}^{i3}\right]+$$
$$\frac{1}{18}m_1c_\gamma g_1^2y_d^{i1i4}\left(-c_\gamma\overline{a_d}^{i2i3}+s_\gamma\tilde\mu\overline{y_d}^{i2i3}\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde q}^{i2},m_1,m_{\tilde d}^{i3}\right]+$$
$$\frac{4}{3}m_3c_\gamma g_3^2y_d^{i1i4}\left(-c_\gamma\overline{a_d}^{i2i3}+s_\gamma\tilde\mu\overline{y_d}^{i2i3}\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde q}^{i2},m_3,m_{\tilde d}^{i3}\right]+$$
$$\frac{4}{3}m_3c_\gamma g_3^2y_d^{i1i4}\left(c_\gamma\overline{a_d}^{i2i3}-s_\gamma\tilde\mu\overline{y_d}^{i2i3}\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde q}^{i2},m_3,m_{\tilde d}^{i3}\right]+$$
$$\frac{1}{2}\tilde\mu c_\gamma\Big(\overline{y_d}^{i2i3}y_d^{pi4}y_u^{i1r}\left(s_\gamma\overline{a_u}^{pr}-\tilde\mu c_\gamma\overline{y_u}^{pr}\right)+\overline{y_d}^{pi3}y_d^{i1i4}\overline{y_u}^{i2r}\left(s_\gamma a_u^{pr}-\tilde\mu c_\gamma y_u^{pr}\right)\Big)$$
$$\mathrm{LF}_{2,1,1,0}\left[m_{\tilde u}^{r},m_{\tilde q}^{p},\tilde\mu\right]+\frac{1}{2}\tilde\mu c_\gamma\Big(\overline{y_d}^{i2i3}y_d^{pi4}y_u^{i1r}\left(-s_\gamma\overline{a_u}^{pr}+\tilde\mu c_\gamma\overline{y_u}^{pr}\right)+$$
$$\overline{y_d}^{pi3}y_d^{i1i4}\overline{y_u}^{i2r}\left(-s_\gamma a_u^{pr}+\tilde\mu c_\gamma y_u^{pr}\right)\Big)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde u}^{r},m_{\tilde q}^{p},\tilde\mu\right]+$$
$$\frac{1}{4}\tilde\mu c_\gamma\Big(\overline{y_d}^{i2i3}y_d^{pi4}y_u^{i1r}\left(-s_\gamma\overline{a_u}^{pr}+\tilde\mu c_\gamma\overline{y_u}^{pr}\right)+\overline{y_d}^{pi3}y_d^{i1i4}\overline{y_u}^{i2r}\left(-s_\gamma a_u^{pr}+\tilde\mu c_\gamma y_u^{pr}\right)\Big)$$
$$\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde u}^{r},\tilde\mu,m_{\tilde q}^{p}\right]+\frac{1}{18}g_1^2g_3^2\mathrm{LF}_{1,1,1,1,-1}\left[m_1,m_3,m_{\tilde d}^{i3},m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{9}m_1m_3g_1^2g_3^2\mathrm{LF}_{1,1,1,1,0}\left[m_1,m_3,m_{\tilde d}^{i3},m_{\tilde q}^{i1}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{18}g_1^2g_3^2\mathrm{LF}_{1,1,1,1,-1}\left[m_1,m_3,m_{\tilde d}^{i4},m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{9}m_1m_3g_1^2g_3^2\mathrm{LF}_{1,1,1,1,0}\left[m_1,m_3,m_{\tilde d}^{i4},m_{\tilde q}^{i2}\right]\delta_{i1i2}\delta_{i3i4}+$$
$$\frac{1}{9}g_1^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,1,-1}\left[m_1,m_{\tilde d}^{i3},m_{\tilde d}^{i4},\tilde\mu\right]+$$
$$\frac{1}{18}g_1^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,1,-1}\left[m_1,m_{\tilde d}^{i3},m_{\tilde q}^{i1},\tilde\mu\right]+$$
$$\frac{1}{18}g_1^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,1,-1}\left[m_1,m_{\tilde d}^{i4},m_{\tilde q}^{i2},\tilde\mu\right]+$$
$$\frac{1}{36}g_1^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,1,-1}\left[m_1,m_{\tilde q}^{i1},m_{\tilde q}^{i2},\tilde\mu\right]+$$
$$\frac{3}{4}g_2^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,1,-1}\left[m_2,m_{\tilde q}^{i1},m_{\tilde q}^{i2},\tilde\mu\right]-$$
$$\frac{1}{6}g_3^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,1,-1}\left[m_3,m_{\tilde d}^{i3},m_{\tilde d}^{i4},\tilde\mu\right]+$$
$$\frac{1}{6}g_3^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,1,-1}\left[m_3,m_{\tilde d}^{i3},m_{\tilde q}^{i1},\tilde\mu\right]+$$
$$\frac{1}{6}g_3^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,1,-1}\left[m_3,m_{\tilde d}^{i4},m_{\tilde q}^{i2},\tilde\mu\right]-$$
$$\frac{1}{6}g_3^2\overline{y_d}^{i2i3}y_d^{i1i4}\mathrm{LF}_{1,1,1,1,-1}\left[m_3,m_{\tilde q}^{i1},m_{\tilde q}^{i2},\tilde\mu\right]\Bigg)$$